And the derivative of the inner function $h(x)$ is

$$\frac{d}{dx}(x^4 + x^3 + 2) = 4x^3 + 3x^2$$

To get the full derivative of the compositional function, we notice that

$$\frac{df}{dx} = \frac{df}{\cancel{dh}} \cdot \frac{\cancel{dh}}{dx}$$

That is, the derivative we want, df/dx, is obtained by multiplying the derivative of the outer function with respect to its input and the inner function (the polynomial) with respect to x.

$$\frac{df}{dx} = \frac{1}{h(x)} \cdot \frac{dh}{dx}$$

$$\frac{df}{dx} = \frac{1}{x^4 + x^3 + 2} \cdot (4x^3 + 3x^2)$$

$$\frac{df}{dx} = \frac{4x^3 + 3x^2}{x^4 + x^3 + 2}$$

You can set this derivative to 0 to calculate the minima algebraically: $4x^2 + 3x = 0$. This function has two minima at $x = 0$ and $x = –3/4 = –0.75$. But only $x = –0.75$ is the global minimum since $f(–0.75) = 0.638971$ whereas $f(0) = 0.693147$, which is slightly larger.

Let’s see how we can solve this using *gradient descent*, which is an iterative algorithm to find the minima of a function. The idea is we start with a random x as a starting point. We then compute the derivative of the function at this point, which tells us the magnitude and direction of curvature at this point. We then choose a new x point based on the old x point, its derivative, and a step-size parameter to control how fast we move. That is,